

DECODELABS

Project 4: Data Visualization & Storytelling

E-Commerce Revenue Performance and Revenue Leakage Analysis

Data Analytics Internship | SkillAhead Solutions Ltd

2023 – 2025 | Power BI

Executive Summary

This project transforms raw e-commerce transaction data into meaningful business insights through data visualization and storytelling using Power BI. The dashboard focuses on two critical business dimensions: revenue performance and revenue leakage. The objective was not simply to visualize data — but to communicate findings that support better strategic decisions.

1. Dataset Overview

Property	Detail
Total Orders	1,200
Total Customers	1,189
Total Products	7
Analysis Period	2023 – 2025
Visualization Tool	Microsoft Power BI

Key Fields:

OrderID, CustomerID, Product, Quantity, TotalPrice, PaymentMethod, ReferralSource, CouponCode, OrderStatus, Date

2. Dashboard Design Approach

Principle	Description
One Page, One Message	Each dashboard page focuses on a single business question
Reduce Visual Clutter	Only visuals that directly support the objective were included
Tell a Story	Every visual answers: What happened? Why does it matter? What next?

3. Dashboard Page 1 — Revenue Driver Analysis

Business Question: What factors are driving revenue growth?

3.1 Key Performance Indicators

\$1,264,762	1,200	1,189	7	5.49
Total Revenue	Total Orders	Total Customers	Total Products	Avg Items in Cart

3.2 Revenue by Referral Source

Referral Source	Total Revenue	Rank
Instagram	\$275,285	#1 — Top Channel
Email	\$261,809	#2
Google	\$250,441	#3
Facebook	\$250,411	#4
Referral	\$226,816	#5

3.3 Revenue by Product

Product	Total Revenue	Rank
Chair	\$195,620	#1
Printer	\$195,613	#2
Laptop	\$192,127	#3
Tablet	\$186,569	#4
Monitor	\$175,651	#5
Desk	\$167,460	#6

Phone	\$151,722	#7
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3.4 Revenue Trend (2023–2025)

Revenue fluctuated throughout the analysis period but remained above \$100K per quarter. Although periodic growth occurred, the overall trendline indicates a gradual decline that may require strategic attention.

3.5 Key Insights — Revenue Drivers

- Insight 1: Instagram is the most effective customer acquisition channel — generating more revenue than any other referral source.
- Insight 2: Revenue is concentrated among a small number of products. Chairs, Printers, and Laptops together account for a significant share of total revenue.
- Insight 3: The business may face future revenue challenges if the declining trendline continues. Growth strategies should be implemented proactively.

3.6 Recommendations — Revenue Drivers

- Increase marketing investment in Instagram campaigns and other high-performing digital channels.
- Maintain adequate inventory levels for top-performing products (Chair, Printer, Laptop).
- Introduce cross-selling and upselling strategies to increase revenue per customer.
- Monitor revenue trends regularly and implement growth initiatives before significant declines occur.

4. Dashboard Page 2 — Revenue Leakage Analysis

Business Question: How much revenue is being lost and what is causing the loss?

4.1 Key Performance Indicators

\$519,674	\$276,396	\$243,278	41.4%
Total Revenue Lost	Lost to Cancellations	Lost to Returns	Combined Loss Rate

4.2 Revenue & Orders by Status

Order Status	Total Revenue	Total Orders	% of Orders
Cancelled	\$276,396	250	20.8%
Pending	\$256,328	237	19.8%

Shipped	\$246,160	235	19.6%
Returned	\$243,278	247	20.6%
Delivered	\$242,600	231	19.3%

4.3 Key Insights — Revenue Leakage

- Insight 1: Order cancellations generate more financial loss (\$276,396) than product returns (\$243,278). Reducing cancellations offers the greatest opportunity for revenue recovery.
- Insight 2: Revenue leakage occurs after customer acquisition costs have already been incurred — meaning the business loses both revenue and marketing investment simultaneously.
- Insight 3: Operational inefficiencies may be contributing to lost revenue. Potential causes include payment issues, checkout abandonment, shipping delays, product dissatisfaction, and customer service challenges.

4.4 Recommendations — Revenue Leakage

- Investigate root causes of order cancellations — focus on checkout process, payment methods, and order confirmation experience.
- Analyze return reasons and identify recurring product or service issues driving returns.
- Implement proactive customer communication throughout the order fulfillment process.
- Develop a revenue recovery strategy focused on reducing both cancellations and returns.

5. Overall Business Insights

Two major business opportunities were identified from this analysis:

Opportunity 1 — Revenue Growth

Revenue performance is strongly influenced by high-performing referral channels and products. Instagram and top products (Chair, Printer, Laptop) should receive increased strategic investment to sustain and grow revenue.

Opportunity 2 — Revenue Recovery

Over \$519,000 in potential revenue was lost through cancellations and returns — representing 41.4% of all order revenue. Reducing revenue leakage may generate substantial financial gains without requiring additional customer acquisition spending.

6. Final Recommendations

1. Increase investment in high-performing marketing channels — particularly Instagram.
2. Prioritize inventory management for top-performing products (Chair, Printer, Laptop).
3. Develop strategies to reverse the declining revenue trend before losses become significant.
4. Investigate and resolve the root causes of order cancellations.
5. Reduce product return rates through quality and customer service improvements.
6. Implement customer retention and engagement initiatives to improve repeat purchase rate.
7. Monitor key business metrics through regular Power BI dashboard reviews.

7. Conclusion

This project demonstrates how data visualization can transform transactional data into actionable business intelligence. By focusing on revenue drivers and revenue leakage, the analysis identified clear opportunities to increase revenue, improve operational efficiency, and support better strategic decision-making.

The findings suggest that future growth can be achieved through stronger investment in successful acquisition channels — particularly Instagram — while simultaneously reducing revenue losses caused by cancellations and returns. Together, these two strategies could significantly improve the business's financial performance over the coming periods.

8. Expected Outcomes

If the recommendations outlined in this report are implemented, the following outcomes are expected across key business areas:

Area	Recommendation	Expected Outcome
Marketing	Increase Instagram investment	Higher customer acquisition and revenue growth from the top-performing channel.
Inventory	Prioritize top product stock levels	Fewer missed sales for Chair, Printer, and Laptop — the three highest-revenue products.
Revenue Growth	Implement upselling & cross-selling	Average order value rises above the current \$1,054 per transaction.
Cancellations	Fix checkout & payment experience	Reduction in the 20.8% cancellation rate and recovery of up to \$276,396 in lost revenue.
Returns	Improve product quality & service	Reduction in the 20.6% return rate and recovery of up to \$243,278 in lost revenue.

Area	Recommendation	Expected Outcome
Revenue Recovery	Execute leakage reduction strategy	Potential recovery of up to \$519,674 without additional customer acquisition spend.
Trend Reversal	Monitor and act on revenue trend	Stabilization and eventual reversal of the current declining revenue trendline.
Retention	Launch customer engagement program	Increased repeat purchase rate from the current near-zero baseline.

Overall Financial Impact Potential

\$519,674	\$275,285+	\$1,054+	< 20%
Recoverable Revenue (Cancellations + Returns)	Growth Potential (Instagram Optimization)	Target Avg Order Value (Upselling Strategy)	Target Cancellation & Return Rate

These outcomes are not guaranteed but represent the realistic financial opportunities identified through the data. The most immediate and high-impact action is addressing revenue leakage — as this requires no additional customer acquisition spend and can deliver measurable results within a short timeframe.

Sign-Off

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Date: June 2026

Data tells a story. This is ours.